# Intermediate flag stabilizers in Geometry 5.1

L100 literal structural supplement

The construction and incidence properties of Geometry 5.1 in the fixed Halle preprint of Gottschalk–Leemans [1] have already been verified on its literal projective model. Its flag orders alone do not prove the intermediate structural names used in Lemma 4.3. This supplement supplies those names, using the acquired ambient permutation list and the acquired structural-certificate lemmas. No subgroup or geometry census is repeated.

Write the source types as $1 = \ell$ (line), $2 = p$ (point), $3 = h_3$, $4 = h_4$ (hyperovals in distinct specified orbits), and $5 = \{p, q\}$. Use precisely the saved construction chamber, whose five objects are the point masks

$$(31, 32, 27808, 2740, 160).$$

These masks refer to the accepted 21-point model, not to untransported coordinate labels or to the differently ordered coset catalogue. For every nonempty $I \subseteq \{1, \dots, 5\}$ define

$$H_I = \{g \in G : \ g \text{ fixes each of these objects of a type in } I\}.$$

The checker computes each set by testing all the acquired 20160 literal permutations, and joins its order to the accepted flag record. Since the acquired construction is flag-transitive, every other flag of the same type has a conjugate stabilizer. Thus a certificate at this chamber pays the structural assertion for every flag of that type.

**Theorem 1.** *All intermediate group names in Lemma 4.3 of the fixed preprint agree with the literal construction. The complete size-two and size-three flag tables are as follows; $D_{10}$ has order ten.*

| $I$ | $H_I$ | $I$ | $H_I$ | $I$ | $H_I$ | $I$ | $H_I$ |
|---|---|---|---|---|---|---|---|
| 12 | $A_5$ | 13 | $A_5$ | 14 | $S_4$ | 15 | $2^3$ |
| 23 | $A_5$ | 24 | $A_5$ | 25 | $2^4 : 3$ | 34 | $3^2 : 2$ |
| 35 | $S_4$ | 45 | $S_4$ | | | | |
| 123 | $D_{10}$ | 124 | $S_3$ | 125 | $2^2$ | 134 | $S_3$ |
| 135 | $2^2$ | 145 | $2^2$ | 234 | $S_3$ | 235 | $A_4$ |
| 245 | $A_4$ | 345 | $S_3$ | | | | |

*Proof.* For each actual set the certificate first gives generators whose literal closure is exactly that set. The acquired parabolic-structure methods then supply faithful alternating actions for $A_5$; conjugation on four actual cyclic-three subgroups for $A_4, S_4$; commuting binary bases for the elementary abelian groups; and complete semidirect normal forms for $S_3$ and $3^2 : 2$. The latter has inversion action on its elementary nine-subgroup and is the same action denoted $3 : S_3$ in the acquired certificate. All witnesses are checked anew on these particular sets; an equal order or an abstract name in another catalogue position is never used as an identification. The two additional methods needed here are established below. □

**Lemma 2.** *The actual $H_{123}$ is $D_{10}$. The actual $H_{25}$ is a split $2^4 : 3$ with fixed-point-free complement action on the nonidentity elements of its normal elementary abelian subgroup.*

*Proof.* In $H_{123}$ the saved elements $a, b$ have orders five and two, respectively, satisfy $a^b = a^{-1}$, and the ten words $a^i b^j$ $(0 \le i < 5,\ 0 \le j < 2)$ are pairwise distinct and exhaust the actual set. These relations and normal forms give mutually inverse homomorphisms with the dihedral group of order ten.

In $H_{25}$ the elements of order at most two form a 16-element subgroup $N$. The saved four generators are commuting involutions and their sixteen binary coordinate words are distinct. Normality is checked by conjugation with the saved generators of $H_{25}$. The saved element $r$ has order three. Its conjugation action on the four basis elements is recorded as an actual matrix over $\mathbb{F}_2$, and it fixes no nonzero coordinate vector. All 48 words $nr^j$, $n \in N$, $0 \le j < 3$, are distinct and exhaust $H_{25}$. Thus $N \cap \langle r \rangle = 1$ and the extension splits with exactly the recorded action. To identify this action abstractly, its minimal polynomial divides $X^3 - 1 = (X-1)(X^2+X+1)$. Absence of fixed vectors eliminates the first factor, so $N$ is the direct sum of two copies of the irreducible two-dimensional $\mathbb{F}_2 C_3$ module. This specifies the semidirect product beyond a mere order comparison. □

The same certificate covers the five singleton stabilizers $2^4 : A_5, 2^4 : A_5, A_6, A_6, 2^4 : S_3$, the five four-object stabilizers (order two for $1234, 1235, 1245, 1345$, and order three for $2345$), and the trivial chamber stabilizer. There are therefore 31 exact nonempty flag types. The singleton structures reuse the acquired geometric stabilizer argument; all cyclic and elementary structures have literal normal forms.

**Verification and source boundary.** `certify.py` retains every element set and structural witness. Its saved-witness validation recomputes the actions and normal forms; damaged generators, action matrices, words, type assignments and source names are rejected. The native source page was read directly. The formula $G_{p,q}$ in its proof refers to the subgroup fixing both points, equivalently the stabilizer of the incident point and the unordered pair. The source's $3^2 : 2$ is normalized to the acquired name $3 : S_3$ only with its explicitly verified inversion action.

This supplement makes no new source correction. Residual connectivity, weak residual primitivity, complete flag extension, the ambient linear realization and flag transitivity are acquired with their prior proofs and distinct reviews. They are not inferred merely from this table of groups. The journal edition and unrelated cited classifications are not recertified here.

## References


[1] H. Gottschalk and D. Leemans, *Geometries for the group* PSL(3, 4), Halle preprint 99–30, fixed 22-page edition, p. 15, Lemmas 4.2–4.3.